Development and Characterisation of Eggshell–Agar Biocomposites for Sustainable Interior Applications



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ICSMMEP-26 · Mecca, Saudi Arabia · 3–4 November 2026

A waste stream in search of a binder

Eggshell is produced as food waste in millions of tonnes each year. It is roughly 94% calcium carbonate, and its porous shell structure is mechanically useful.

The shells for this study came from one bakery in Ankara, collected over more than a year.



Composition of eggshell

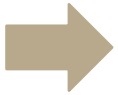
94%	calcium carbonate
4%	organic matter
1%	calcium phosphate
1%	magnesium carbonate

The filler was already bio-based. The binder was not

Eggshell in the literature

- Cement and concrete replacement
- Asphalt binder modification
- Soil stabilisation, replacing lime
- Ceramic tile and wallpaper production
- Filler in PLA, epoxy, PVC and polyurethane
- Reinforcement in metal and ceramic composites
- Bio-calcium carbonate for wall tiles and paper

In every case the matrix stays petrochemical or clinker-based.



This study

Filler: waste eggshell powder

Binder: agar and glycerol

Finish: shellac

Both phases are biological and biodegradable.

Four components, all biodegradable

Eggshell powder

Mineral phase. Sieved to 150 µm.

Agar

Seaweed polysaccharide. Thermoreversible gel.

Glycerol

Plasticiser. Governs ductility.

Water

Carrier. Governs final density.



Prototype variations in colour, texture and format. Natural pigments including beetroot were trialled.

Why the protocol is the contribution

1

No shared procedures exist

Designer-led biofabrication has no standard protocol. Each study starts from scratch.

2

So the process stays trial and error

Finding a workable formulation takes months of iteration that nobody else can reuse.

3

And results cannot be compared

Without recorded parameters, two studies of the same material are not comparable at all.

This paper reports every parameter needed to reproduce the material.



Cast panel, 300 mm square

Preparing the shell fraction

1. **Clean**
   Membrane and organic residue removed

2. **Heat treat**
   120 °C for 20 minutes, for decontamination and drying

3. **Cool and dry**
   To room temperature, then dried until no residual moisture

4. **Mill**
   Laboratory mechanical grinder, until visually homogeneous

5. **Sieve**
   150 µm mesh; only the passing fraction is used



120 °C
heat treatment

150 µm
sieve aperture

Preparing the binder phase

1. **Combine**
   Water and glycerol, stirred until homogeneous

2. **Add agar**
   Introduced gradually under continuous stirring to prevent agglomeration

3. **Dissolve**
   Heated to 95 °C until the agar is fully in solution

4. **Hold**
   Reduced to 80 °C and cooked at that temperature, stirring throughout

5. **Add the mineral**
   Shell fraction added incrementally near the end of cooking

Key parameters

95 °C
agar dissolution

80 °C
controlled cooking

55 min
total heating time

Stirring was manual throughout, so shear rate is not controlled. This is stated as a limitation.

Casting, drying and consolidation

1. **Cast**
   Poured into silicone moulds and spread to uniform thickness. Silicone flexes, so no release agent is needed.

2. **Set in mould**
   Four days for initial setting and drying

3. **Demould**
   Removed once the specimen holds its form

4. **Consolidate**
   A further seven days under ambient conditions, shielded from direct sun and air movement

Dimensional loss

The material shrinks appreciably as the water phase evaporates. Post-drying dimensions were not recorded in this study; the dimensions reported here are mould dimensions. Future work using this protocol should measure specimens after drying, not at casting.

23 °C
ambient drying

11 days
mould to finished

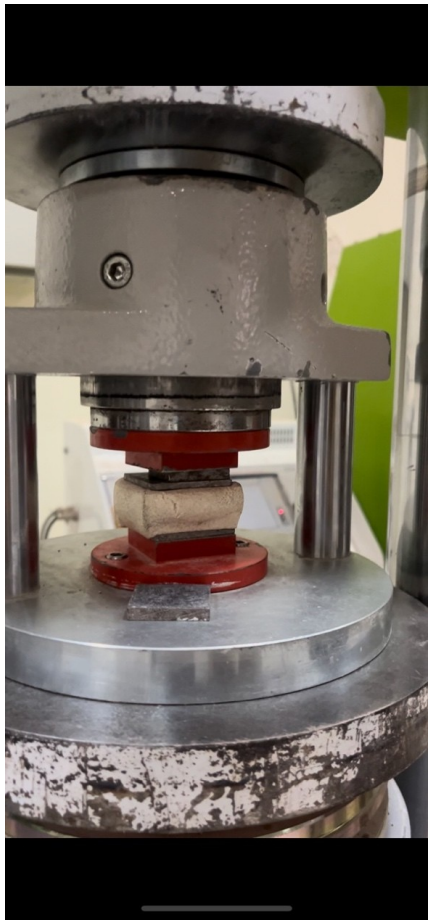
Formulations characterised

Test	ID	Water (g)	Eggshell (g)	Agar (g)	Glycerol (g)	Outcome
Compression	C1	325	80	35	5	2.55 MPa
Compression	C2	310	90	40	5	3.12 MPa
Compression	C3	280	110	45	10	**4.10 MPa**
Flexure	F1	450	190	70	60	*not measurable*
Flexure	F2	480	190	70	30	**2.33 MPa**

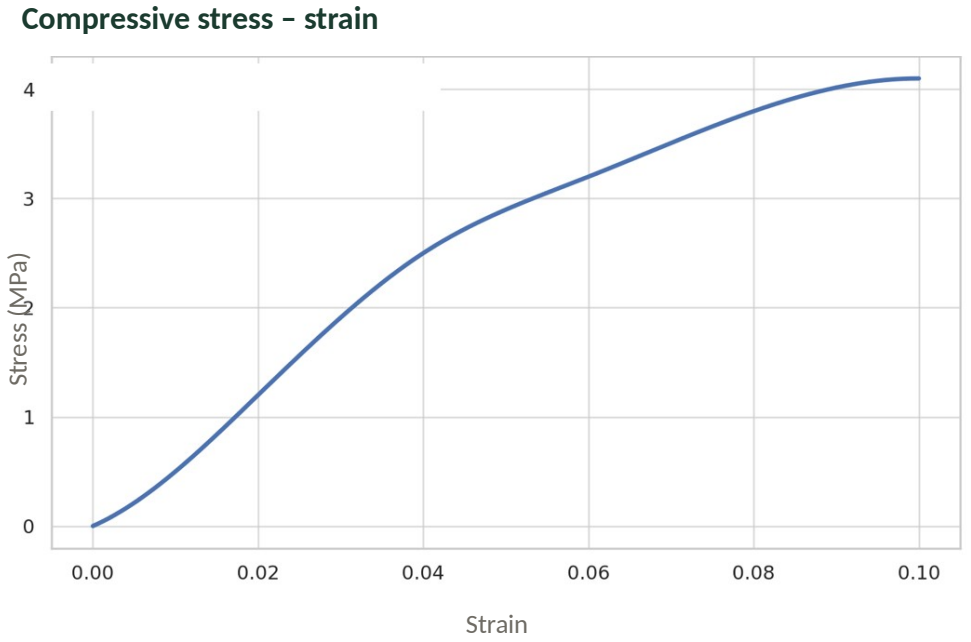
Three specimens were tested per formulation. Values reported are means; individual specimen values were not retained, so standard deviations are not available.

Two relationships hold across the set: less water with more mineral and agar raises load capacity; halving glycerol turns an unmeasurably ductile material into a measurable one.

Compression: it crushes without disintegrating



Barrelled specimen, still in one piece



Past peak load the stress does not collapse. The specimen keeps deforming while holding together, which is energy absorption rather than fracture.

4.10 MPa
mean compressive strength, C3

6.80 kN
maximum load at failure

~1.4 g/cm³
density, below brick and concrete

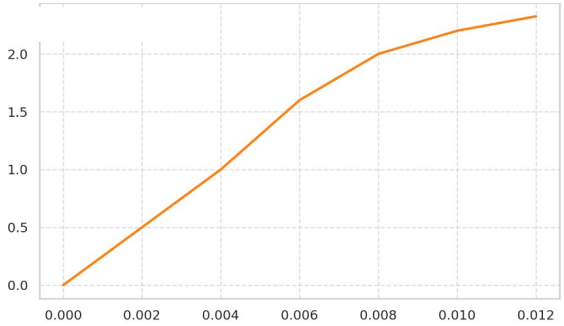
Flexure: glycerol governs everything





Kerfed panel arched by hand

Flexural stress – strain (MPa / strain)



2.33 MPa
flexural strength, F2

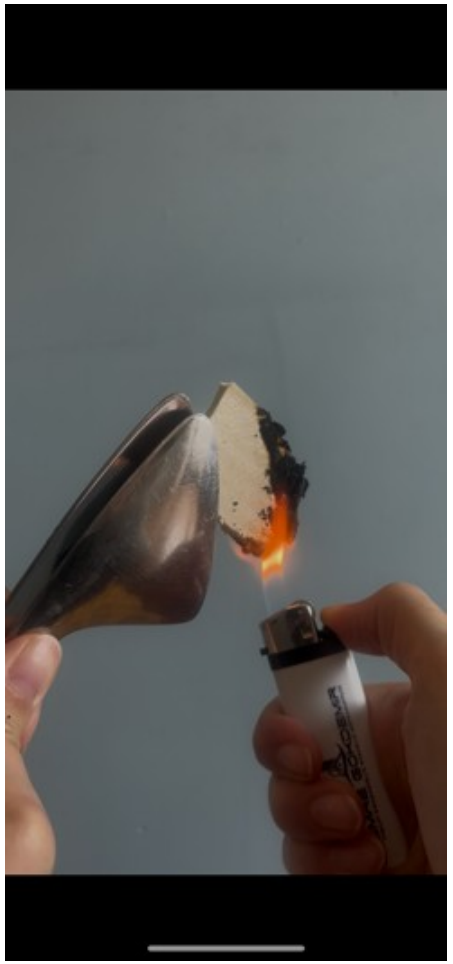
192 MPa
flexural modulus

At 60 g glycerol the specimen was too ductile to register a reading. At 30 g it became measurable while still flexing.

A note on the cross-section

The specimen carries transverse kerfs on its tension face, visible in both photographs. ASTM D790 assumes a solid rectangular section, and the calculation here uses the full 60 × 20 mm. Part of the measured compliance is therefore geometric, not material.

Fire and water response



Single-flame ignitability

ISO 11925-1, 15 s flame at 45°

~10 s

to ignition

4 s

to self-extinguish after flame removal

Charring stayed local. No flame spread, no melting, no dripping.

Water absorption

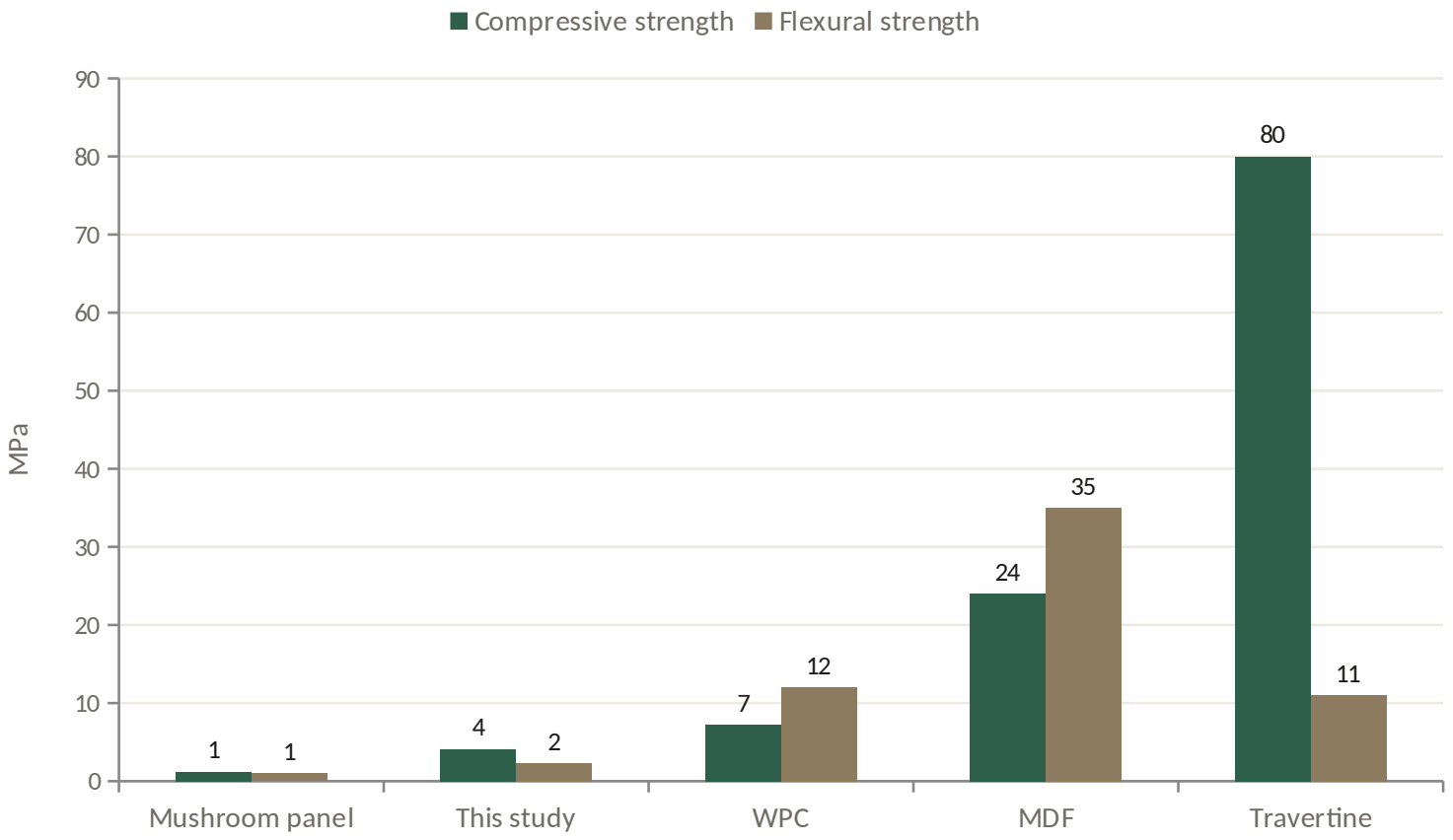
ASTM D570, 24 h full immersion

0.7%

mass gain, shellac-coated

Shellac is a natural, biodegradable, hydrophobic resin, applied here in three coats. Where the coating was thin, surface loss occurred. Coating integrity is the limiting factor for any humid application.

Judged by strength, this material is weak



On the conventional criterion the conclusion is simple.

It is one sixth the compressive strength of MDF and one twentieth that of travertine — a stone it happens to resemble.

So we stop here, and the material is filed as not strong enough.

Sources: Velasco & Gaona 2011; Klyosov 2007; Nemli & Aydın 2007; Altun 2006

But strength is not the only thing worth measuring

Modulus of resilience

$$Ur = \sigma f^2 / 2E$$

The elastic energy a material absorbs per unit volume before it takes permanent damage.

A low stiffness raises this value. The property reported as a weakness is what produces it.

$$2.33^2 / (2 \times 192) = 14.1\ kJ/m^3$$

for the formulation measured here





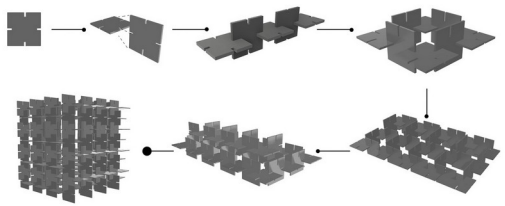
Neither specimen fractured. Both absorbed load and kept their integrity — the behaviour strength figures do not capture.

The proposal

Report this value for bio-based materials. Almost no study currently does, so the comparative set does not yet exist.

Where a material like this belongs





Non-loadbearing surfaces

Panels, cladding and partitions, where stiffness is not the governing requirement

Modular and demountable systems

Components designed to be reconfigured rather than to last unchanged

Temporary and exhibition structures

Short service life, where biodegradability is an advantage, not a defect

Modular interior proposal developed from the material's measured properties, from the author's doctoral research.

What this study does not establish

Three specimens, means only

Individual specimen values were not retained, so no standard deviations are reported.

Shrinkage not recorded

Dimensions quoted are mould dimensions; the dried cross-section was measured at test time but not logged.

Kerfed section treated as solid

The ASTM D790 calculation assumes a full rectangular section, so some compliance is geometric.

No long-term data

Humidity cycling, temperature cycling and microbial resistance were not tested.

One laboratory, one operator

Stirring was manual; the process has not been reproduced elsewhere.

Prototype scale only

The design proposal was never built at full scale or occupied.

Stating these is part of the contribution: a protocol that others are meant to repeat has to say where it is thin.

01

A complete protocol

Every parameter needed to reproduce an eggshell–agar biocomposite is reported.

02

Measured, with limits stated

4.10 MPa compressive, 2.33 MPa flexural, self-extinguishing, 0.7% water uptake.

03

A criterion worth changing

Deformation capacity, not strength alone, is what makes this class of material useful.

Thank you

Questions welcome

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With thanks to the Advanced Construction Materials Research Laboratory, Hacettepe University, and to Dr Hüseyin İlcan. Derived from the author's doctoral dissertation, Hacettepe University, 2025.